Crypto Due Diligence Analysis

Investment memo framework, source standards, and student template

Educational use only. This assignment is designed for academic due diligence and is not financial advice.

Assignment: Crypto Due Diligence Investment Memo

Choose one crypto asset, token, blockchain network, DeFi protocol, or blockchain venture. Prepare an MBA-level due diligence memo that answers a single decision question: should an informed investor or strategic partner buy, hold, avoid, or watch this asset? Your memo must evaluate the project, the token, the market, the technology, the economics, the valuation, and the risks. You must distinguish between project quality and token investment quality. A project can be useful while its token remains a poor investment.

Your memo/ppt must include:

- Executive summary and final recommendation.
- Project background, team, funding, and credibility.
- Market, customer problem, use case, adoption, and competition.
- Technology, security, scalability, and decentralization.
- Tokenomics, value capture, supply, allocation, emissions, staking, burns, and unlocks.
- Governance, treasury control, voting power, and insider incentives.
- Valuation using market cap, FDV, relevant ratios, comparables, and bear-base-bull scenarios.
- Liquidity and investor exit analysis.
- Regulatory, tax, operational, cybersecurity, and reputational risks.
- Disconfirming evidence and monitoring plan.

Students must include ppt slides: project snapshot, market map, token unlock chart, value capture diagram, valuation comparables, governance map, risk heatmap, and final recommendation page. Use credible sources and provide a source log. Project marketing materials are allowed, but they must be verified against independent sources, market data, and on-chain data where possible.

Teams, Categories and Candidate Coins

The class works in five teams of four students. Each team investigates a different category and one cryptocurrency from it: no two teams share a category or a coin. The last column suggests a useful due-diligence comparison within the category.

Category	Candidate coins	A useful due-diligence comparison
AI	Bittensor (TAO), Render (RENDER), Artificial Superintelligence Alliance (FET), Virtuals Protocol (VIRTUAL), Pearl Research (PRL), Perle Labs (PRL)	Compare AI service markets, GPU compute and expert data work. Test whether there are paying customers, how work quality is verified, whether token demand follows real usage and whether emissions overwhelm demand. Bittensor's subnet incentives, Render's burn-and-credit model and Pearl's proposed proof of useful work are quite different token-economy cases (bittensor.com).
Gaming and game assets	Sui (SUI), Solana (SOL), Aptos (APT), Immutable (IMX), Ronin (RON), Axie Infinity (AXS)	Compare general-purpose platforms with networks built around game distribution and assets. Ask whether games retain active players, whether on-chain items improve the player experience and whether activity creates value for the network token. Sui, Ronin and Immutable all provide game-development infrastructure, but their tokens have different roles (docs.sui.io).
Decentralized storage and cloud infrastructure	Filecoin (FIL), Internet Computer (ICP), BitTorrent (BTT), Arweave (AR), Akash (AKT)	Compare persistent storage, web infrastructure and decentralized cloud markets. Test paid demand, retrieval performance, provider concentration, customer retention and whether token rewards are needed to make the service economical. Filecoin, for example, uses storage deals and provider collateral

		(docs.filecoin.io).
DeFi lending and decentralized exchanges	Aave (AAVE), Morpho (MORPHO), Uniswap (UNI), Curve (CRV), Aerodrome (AERO), PancakeSwap (CAKE)	Compare lending markets with token-swapping venues. Investigate who bears smart-contract and collateral risk, how the protocol attracts liquidity and whether fees or governance rights create value for token holders.
On-chain trading and derivatives	Hyperliquid (HYPE), Injective (INJ), Jupiter (JUP), dYdX (DYDX)	Compare an on-chain derivatives venue, a trading-focused chain and a trading aggregator. Examine volume quality, fee distribution, leverage and liquidation risks, token unlocks and how much activity depends on incentives. Hyperliquid's published fee mechanics make it a useful case for tracing protocol fees to token value capture (hyperliquid.gitbook.io).
Tokenized real-world assets and commodities	Ondo (ONDO), PAX Gold (PAXG), Tether Gold (XAUt), Centrifuge (CFG), Maple (SYRUP)	Compare tokenized securities, gold-linked tokens and on-chain credit. Ask what legal rights token holders actually have, how redemption works, who holds the underlying assets and whether the project token captures value from the products. Ondo's own terms distinguish its tokens from the underlying securities (app.ondo.finance).
Privacy and confidential transactions	Monero (XMR), Zcash (ZEC), Dash (DASH)	Compare privacy by default with optional privacy features. Examine the cryptographic design, usability, auditability, exchange access and regulatory exposure.
Oracles and interoperability	Chainlink (LINK), Pyth Network (PYTH), LayerZero (ZRO), Quant (QNT), Flare (FLR)	Compare data feeds with cross-chain messaging and interoperability. Test security assumptions, integration concentration, actual fees and what happens when an oracle or bridge fails.
Payments and monetary networks	Bitcoin (BTC), XRP (XRP), Stellar (XLM), Litecoin (LTC), Bitcoin Cash (BCH), Kaspa (KAS)	Compare monetary design, payment use, consensus and security budgets. Kaspa adds a different blockDAG approach, so you can test technical claims against security and adoption evidence (wiki.kaspa.org).
Exchange-linked tokens	BNB (BNB), Cronos (CRO), OKB (OKB), KuCoin Token (KCS), GateToken (GT), Bitget Token (BGB)	Compare tokens linked to centralized exchange businesses. Examine the token's enforceable utility, dependence on the exchange, transparency, related-party risks and what happens to holders if the business faces regulatory or operational trouble.
Meme and community-driven assets	Dogecoin (DOGE), Shiba Inu (SHIB), Pepe (PEPE), Pudgy Penguins (PENGU), OFFICIAL TRUMP (TRUMP)	Use these to analyze attention, community, distribution and speculative demand. Ask what, if anything, supports ongoing value beyond brand recognition and trading interest.

Suggested Data and Research Resources

Category	Examples
Market and supply data	CoinGecko, CoinMarketCap, Messari, Token Terminal, DeFiLlama.
On-chain analytics	Etherscan, blockchain explorers, Dune, Nansen, Glassnode, Santiment.
Security	Audit reports from reputable firms, Immunefi, bug bounty pages, exploit databases, governance forums.
Legal and regulation	Regulator websites, court filings, enforcement releases, reputable legal analyses.
News and research	The Block, CoinDesk, CoinTelegraph, academic articles, credible industry research.
Community evidence	Governance forums, Discord, Telegram, Reddit, X. Use cautiously and verify.

Step	Core question	MBA output
1. Screen	What is this asset, what does it claim to solve, and why now?	One-page project snapshot and initial thesis.
2. Validate	Is there evidence of real users, real activity, credible technology, and credible people?	Evidence-backed operating analysis.
3. Analyze economics	Does the business model create value, and does the token capture that value?	Tokenomics and value capture analysis.
4. Value	What is the asset worth under reasonable scenarios?	Comparable valuation, ratio analysis, and scenario table.
5. Stress test	What can go wrong, and what evidence would falsify the thesis?	Risk heatmap, red flags, and disconfirming evidence.
6. Decide	Buy, hold, avoid, or watch, and under what conditions?	Investment memo and recommendation.

Student Assignment Brief

Task. Choose one crypto asset, token, blockchain network, DeFi protocol, or crypto infrastructure project. Prepare a professional due diligence memo that evaluates whether a sophisticated investor or strategic partner should buy, hold, avoid, or watch the asset. Your analysis must be evidence-based, comparative, and decision-oriented.

Required deliverables. Present a 10 minute boardroom-style summary with 10-12 slides. Use the language of due diligence, not promotion.

Required final decision. Your final answer must be one of the following: Buy, Hold, Avoid, or Watch. If the project has no investable token, use Partner, Do Not Partner, or Watch.

Important limitation. This is an academic exercise. It is not financial advice. Students should not rely on project marketing materials as sufficient evidence.

Learning Objectives

- Separate project quality from token investment quality.
- Evaluate tokenomics, incentives, governance, and value capture.
- Apply valuation tools to crypto assets despite uncertainty and incomplete disclosure.
- Use source triangulation and skeptical evidence standards.
- Build a defensible investment memo under uncertainty.
- Identify red flags, downside risks, and evidence that would change the thesis.

Due Diligence Framework

1. Executive Summary and Investment Thesis

- What is the project in one sentence?
- What is the asset being evaluated: L1 coin, L2 token, DeFi governance token, utility token, stablecoin, NFT, infrastructure token, or equity-like claim?
- What is the recommendation: Buy, Hold, Avoid, or Watch?
- What is the thesis in three bullets?
- What are the top three risks?
- What new evidence would change the recommendation?
- What is the expected time horizon for the thesis?

2. Project Background and Credibility

- Creation date, whitepaper date, launch date, token generation event, and major roadmap milestones.
- Founders, technical leads, advisors, board or foundation members, prior exits, prior failures, and reputational record.
- Funding history: VC, angels, foundation treasury, grants, ICO, IDO, IEO, airdrops, strategic investors, and ecosystem funds.
- Legal entity, foundation structure, jurisdiction, and disclosure quality.
- Key question: does the team have the ability and incentives to execute over the next 3 to 5 years?

3. Market, Use Case, and Problem Fit

- What customer problem is being solved?
- Is the problem economically important or mostly narrative-driven?
- Who pays, who uses, who benefits, and who captures value?
- Market size, growth rate, adoption barriers, and regulatory barriers.
- Comparison to centralized alternatives and non-token blockchain alternatives.
- Evidence of product-market fit: active users, transaction count, retention, revenue, integrations, developer adoption, or enterprise use.

4. Technology and Security

- Consensus mechanism, architecture, smart contract design, scalability, throughput, finality, interoperability, and dependency on bridges or oracles.
- Audits, bug bounties, exploit history, formal verification, admin keys, upgradeability, and emergency controls.
- Node or validator structure, Nakamoto coefficient or equivalent decentralization proxy, client diversity, and infrastructure concentration.
- Key question: is the technical design necessary for the business model, or is blockchain being used mainly for fundraising and narrative?

5. Tokenomics and Value Capture

- Token supply: circulating supply, total supply, max supply, inflation, emissions, burn mechanisms, and monetary policy.
- Initial allocation: team, investors, foundation, community, treasury, ecosystem incentives, liquidity mining, and airdrops.
- Unlock schedule: cliffs, vesting, 3 month, 12 month, and 24 month unlock pressure.
- Token utility: payment, access, staking, governance, collateral, fee discount, settlement, security, or revenue participation.
- Value capture: does increased network usage increase token demand, fees, burns, staking rewards, or governance value?
- Real yield versus inflation-funded yield: identify whether staking rewards come from revenue or newly issued tokens.
- Key question: does the token need to exist, and does it capture economic value?

6. Governance and Control

- Who controls upgrades, treasury, parameter changes, emissions, grants, and emergency actions?
- DAO voting structure, quorum, delegate concentration, whale influence, governance attack risk, and multisig signers.
- Foundation or company control versus community control.
- Conflicts between token holders, founders, validators, users, liquidity providers, and VCs.
- Key question: if token holders have governance rights, do those rights protect value or create new risks?

7. Adoption, Traction, and Network Effects

- Active wallets, daily or monthly active users, transaction volume, fees, revenue, TVL, developer activity, GitHub activity, integrations, and ecosystem projects.
- User quality: organic users versus mercenary yield farmers, bots, wash activity, or airdrop hunters.
- Network effects: liquidity network effects, developer network effects, data network effects, brand or community network effects, and switching costs.
- Key question: is growth durable after incentives decline?

8. Competition and Strategic Positioning

- Direct competitors, substitutes, centralized alternatives, and emerging technologies.
- Relative advantages: speed, cost, security, liquidity, composability, brand, developer community, regulation, or partnerships.
- Relative weaknesses: technical debt, liquidity fragmentation, governance complexity, fees, or regulatory risk.
- Moat analysis: what prevents users, liquidity, developers, and validators from leaving?

9. Valuation and Comparables

- Market capitalization, fully diluted valuation, circulating supply, liquidity, exchange listings, and trading volume quality.

- Relevant ratios: FDV to revenue, market cap to fees, FDV to TVL, market cap to active users, NVT, and fee yield where applicable.
- Comparable set: at least 3 to 5 comparable protocols or assets.
- Scenario analysis: bear, base, and bull cases with assumptions stated clearly.
- Sensitivity analysis: token price impact of supply growth, user growth, fee growth, token burn, and valuation multiple changes.
- Key question: what expectations are already priced in?

10. Liquidity and Investor Exit

- Where can investors buy and sell the token?
- Depth and quality of liquidity across reputable centralized and decentralized exchanges.
- Bid-ask spread, slippage, trading concentration, wash trading concerns, and market maker dependence.
- Large holder concentration and insider unlock risk.
- Key question: can an investor realistically exit without moving the market?

11. Legal, Regulatory, Tax, and Compliance Risk

- Potential classification as security, commodity, payment token, governance token, or regulated financial product.
- KYC, AML, sanctions, stablecoin, DeFi, derivatives, consumer protection, and market abuse risk.
- Jurisdiction of issuer, foundation, users, validators, and exchanges.
- Regulatory catalysts: ETF approval, enforcement action, licensing, bans, or MiCA-type frameworks.
- Key question: could regulation impair trading, usage, listings, or token economics?

12. Risk, Red Flags, and Thesis Stress Test

- Technical risks: exploits, bridges, oracles, smart contract bugs, centralization, and network outages.
- Economic risks: inflation, low utility, weak value capture, unsustainable incentives, low liquidity, and high FDV.
- Governance risks: insider control, voter apathy, governance attacks, or treasury mismanagement.
- Market risks: crypto cycle, BTC correlation, liquidity shocks, and risk-on sentiment.
- Reputational risks: founder misconduct, misleading claims, fake partnerships, community toxicity, or regulatory warnings.
- Disconfirming evidence: list the 3 facts that would make your recommendation wrong.

Valuation Metrics and Formulas

Metric	Formula or definition	How to interpret
Market capitalization	Token price x circulating supply	Current market value of circulating tokens.
Fully diluted valuation	Token price x maximum or total supply	Value if all tokens were circulating. High FDV with low float can signal future sell pressure.
Float ratio	Circulating supply / total supply	Low float can inflate price and create unlock risk.
Inflation rate	New tokens issued over period / circulating supply	High inflation requires strong demand growth to avoid dilution.
12 month unlock pressure	Tokens unlocking in next 12 months / circulating supply	A measure of near-term supply pressure.
Market cap to TVL	Market cap / total value locked	Useful for DeFi, but TVL quality must be assessed.
FDV to fees or revenue	FDV / annualized protocol fees or revenue	Comparable to revenue multiples, but only useful when fees are durable and captured.
Fee yield	Annualized protocol revenue to tokenholders / market cap	Only meaningful when tokenholders actually receive or capture fees.
NVT	Network value / transaction volume	A rough measure of value relative to network activity.
Token velocity	Transaction volume / market cap	High velocity may weaken token value capture unless offset by staking, collateral use, burns, or holding demand.

Red Flag Checklist

Area	Red flag	Why it matters
Team	Anonymous or unverifiable team with high treasury control.	Accountability and execution risk.
Disclosure	Whitepaper vague, outdated, or mostly marketing.	Weak evidence base and higher uncertainty.
Tokenomics	Low circulating float and high FDV.	Future unlocks may create severe sell pressure.
Insiders	Large insider allocation with short vesting.	Misalignment and pump-and-dump risk.
Staking	High yield funded mainly by emissions.	Yield may be dilution disguised as income.
Utility	Token is not needed for the product to function.	Weak value capture and regulatory risk.
Governance	Voting controlled by team, VCs, or a few whales.	Tokenholder governance may be symbolic.
Security	No credible audits or bug bounty for smart contracts.	Higher exploit risk.
Liquidity	Trading volume concentrated on low-quality exchanges.	Exit risk and possible wash trading.
Adoption	Usage disappears when incentives fall.	Mercenary capital rather than real demand.
Partnerships	Unverified or exaggerated partnership claims.	Marketing risk and credibility risk.
Regulation	Core product resembles unlicensed securities, lending, derivatives, or payments activity.	Listings, access, and business model may be impaired.

Evidence and Source Standards

triangulate sources. Students should separate project claims from independent evidence and from market data. Every important claim should be supported by at least two source types when possible.

Source level	Examples	Use in memo
Primary project sources	Whitepaper, docs, GitHub, tokenomics page, governance forum, foundation reports.	Good for claims, design, roadmap, allocations, and official disclosures. Not sufficient alone.
On-chain and market data	Block explorers, DeFiLlama, Token Terminal, Dune, Nansen, Glassnode, Santiment, CoinGecko, CoinMarketCap.	Use for activity, supply, holders, TVL, fees, liquidity, and adoption.
Independent research	Messari, The Block, Delphi, academic papers, audit firms, regulator publications.	Use for context, criticism, market sizing, and risk assessment.
Community and social sources	X, Discord, Telegram, Reddit, governance comments.	Use cautiously for sentiment, controversy, and community health. Do not treat as verified facts.
News and legal sources	Reputable financial media, court filings, agency statements, enforcement actions.	Use for regulatory, reputational, and event-driven risk.

Optional Scorecard for Investment Committee Discussion

Dimension	Score 1	Score 3	Score 5
Problem importance	Weak or unclear problem.	Useful but crowded or niche problem.	Large urgent problem with strong willingness to pay.
Team credibility	Unverified or weak execution record.	Competent but limited track record.	Strong technical and commercial track record.
Technology	Unproven, insecure, or unnecessary.	Functional with some risks.	Differentiated, secure, and necessary.
Adoption	Mostly hype or incentives.	Some real usage.	Durable organic usage and retention.
Token utility	Token not needed.	Some utility but weak capture.	Token is essential and captures value.
Token supply design	High dilution and insider sell risk.	Manageable but imperfect.	Transparent, aligned, and sustainable.
Valuation	Expensive with weak fundamentals.	Reasonable under base case.	Attractive risk-adjusted upside.
Governance	Centralized or opaque.	Mixed decentralization.	Transparent, accountable, and resilient.
Liquidity	Thin, poor-quality markets.	Adequate but concentrated.	Deep, reputable, diversified liquidity.
Regulatory risk	Business model likely impaired.	Manageable but uncertain.	Low or well-mitigated regulatory exposure.

Suggested interpretation: 40 to 50 = strong candidate, 30 to 39 = watch or hold, below 30 = avoid unless there is a specific catalyst or strategic reason.

Exhibits to Consider

Exhibit	Purpose
1. Project snapshot	Asset type, launch date, chain, market cap, FDV, supply, exchanges, and recommendation.

2. Market map	Show competitors, substitutes, users, and ecosystem position.
3. Token supply and unlock chart	Show current supply, future unlocks, insider allocation, and emissions.
4. Value capture diagram	Show who pays, where fees go, how the token accrues value, and leakage points.
5. Comparable valuation table	Compare market cap, FDV, TVL, revenue, fees, users, and multiples.
6. Governance and control map	Show foundation, DAO, multisig, validators, team, VC holdings, and treasury control.
7. Risk heatmap	Rank risks by likelihood and severity.
8. Final recommendation page	Summarize thesis, upside, downside, catalysts, and monitoring plan.

Student Memo/PPT Template

- Title: Due Diligence Memo on [Project/Token Name]
- Recommendation: Buy, Hold, Avoid
- 1. Executive summary: one page maximum.
- 2. Project overview: what it does and why it matters.
- 3. Market and competitive analysis.
- 4. Technology and security analysis.
- 5. Tokenomics and value capture.
- 6. Valuation and scenarios.
- 7. Governance, liquidity, and exit.
- 8. Regulatory and operational risks.
- 9. Final recommendation and monitoring plan.
- Appendices: exhibits, source log, calculations, and data tables.

Suggested Classroom Discussion Questions

- Is the project valuable even if the token is not?
- What would make this project investable for an institutional investor?
- Which risk is most likely to destroy value: technology, governance, regulation, liquidity, or tokenomics?
- Would the project work better with equity, a stablecoin, a utility token, or no token at all?
- What evidence would prove that adoption is real and not incentive-driven?
- Does the token create fair incentives for users, founders, validators, investors, and developers?

Example Final Recommendation Table

Decision item	Student answer
Recommendation	Buy, Hold, Avoid
Time horizon	Example: 12 to 24 months.
Core thesis	Three concise reasons the asset should outperform, underperform, or be monitored.
Main upside catalyst	Example: fee switch, major integration, regulatory clarity, ecosystem growth, or security upgrade.
Main downside catalyst	Example: token unlock, exploit, governance capture, regulatory action, or user decline.
Valuation view	Cheap, fair, expensive, or not meaningfully valuable with available data.
Confidence level	High, medium, or low, with reason.
Disconfirming evidence	Three facts that would change the recommendation.